by the timing gear and chain the cam shaft rotates. By the rotation of cam shaft the tappet started to move upwards. As the tappet is directly in contact with the valve stem when the tappet moves upwards the valve stem due to the lift of the tappet moves upwards and opens the port. By repeating the cam shaft rotation and tappet movement both inlet and exhaust valves open and close for inducting the charge and expelling the hot gases.

5.3.14 Manifold

Manifolds are the passages through which the air or air fuel mixture enters into the combustion chamber and exhaust gases from the combustion chamber are expelled out. The manifold are of two types they are,

a) Inlet Manifold

b) Exhaust Manifold

a) Inlet Manifold

Inlet manifold is used to pass the air from the filter of the diesel engine or air fuel mixture from carburetor of the petrol engine to the combustion chambers of all the cylinders. This intake manifold is generally made of cast iron. This is fitted at the top of the cylinder head.

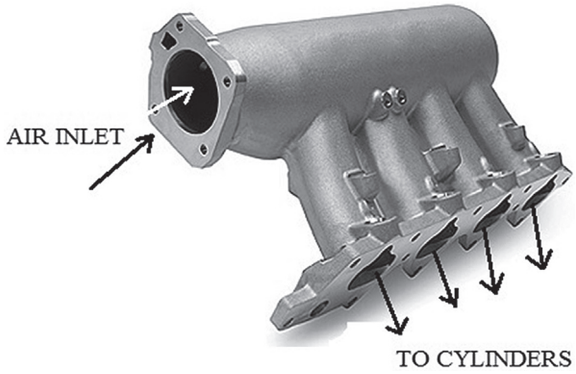


Figure 5.3.14(a) Inlet Manifold

b) Exhaust Manifold

This exhaust manifold is used to transmit the exhaust gases from the cylinder to the silencer. This is fitted between the cylinder head and silencer. This is made of cast iron.

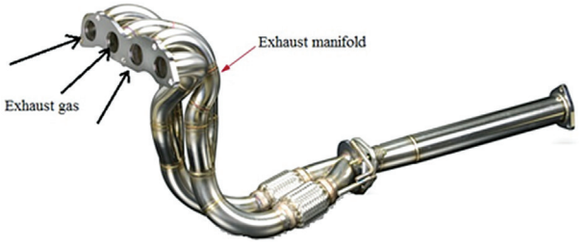
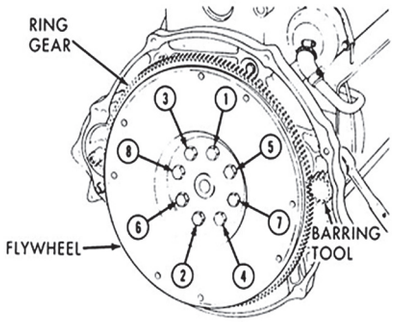


Figure 5.3.14(b) Exhaust Manifold

5.3.15 Flywheel

This is the important component of the engine. This arrangement is used for storing the energy and releasing the energy. It is used to rotate the crankshaft continuously from the initial condition due to its inertia. This is fitted at the end of the crankshaft. It



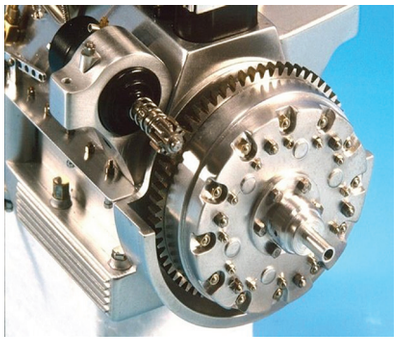


Figure 5.3.15 Flywheel

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